

- (3) a. *The big dog.*
 b. **The dog big.*
 c. *The big cat.*

Other combinations of multiple errors in word form and position are handled in a similar manner. Of course, if the sentences have too little agreement, and therefore no consensus, averaging is unlikely to find any valid solution.

4. RESULTS.

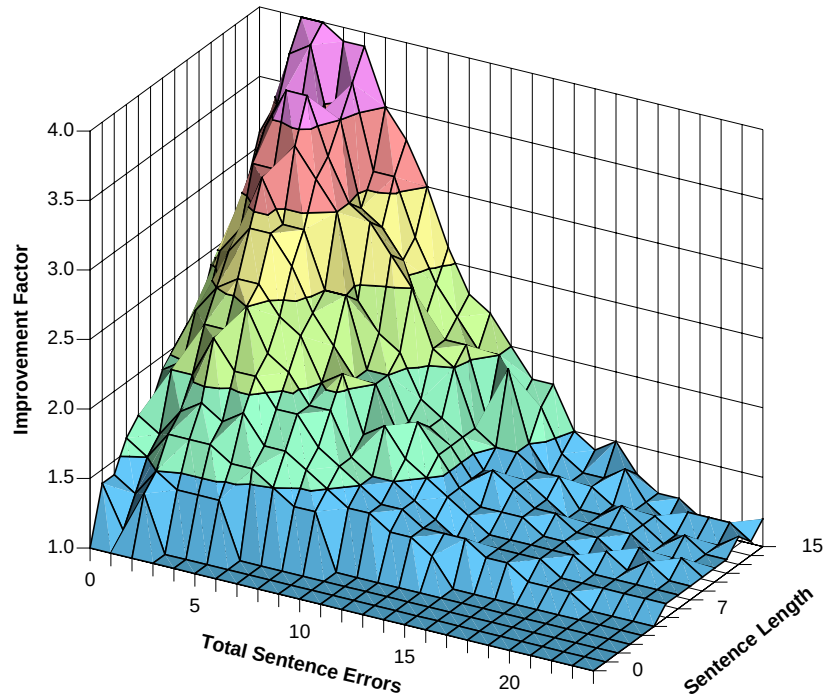
The true test of sentence averaging is how well it performs on actual output from MT packages. However, having the packages produce the test sentences makes it difficult and unwieldy to control the wide range of test parameters. Instead, for the results here, a flexible test generator was employed. For each test, the generator created a group of three sentences S_{1-3} of equal length with 1 to 15 words each and 0 to 120 total random errors between them. These sentences correspond to translations *a*, *b*, and *c* in Figure 1, respectively. A fourth sentence S_{correct} of equal length was generated with no errors. As the correct sentence, its role was to evaluate the quality of S_{1-3} . It has no correspondence in Figure 1 because it was available only in the test environment; with actual translations, correctness cannot be determined automatically.

Since consensus-based averaging relies purely on comparing which words appear in which positions, the actual English grammaticality of S_{1-3} and S_{correct} was not considered. Grammaticality was instead relative only to S_{correct} . The test generator was therefore free to create any sequence of words from its limited lexicon. Furthermore, since straightforward pattern matching is employed for comparison, the test generator and the averaging module are entirely language independent. For example, German sentences could be averaged as easily as sentences containing arbitrary symbols (e.g., ‘A’, ‘B’, etc.).

S_{1-3} were each compared against S_{correct} to determine their correctness. The numerical average C_{group} of these three values reflected the correctness of the entire group. S_{1-3} were then fed through the averaging module to generate the average sentence S_{average} . It was also compared against S_{correct} to determine its correctness C_{average} . The difference between C_{average} and C_{group} indicated the relative improvement: a factor of two, for example, meant that S_{average} was roughly twice as correct (i.e., had half as many errors).

For each of the 1,800 combinations of word length and total error count, 1,000 random tests were run, and an average improvement factor was recorded. The offset conical shape of the results in Figure 2 illustrates an obvious relationship: longer sentences with fewer errors average better than shorter sentences with more errors. This performance is understandable because having more errors reduces the likelihood of consensus between S_{1-3} . The best performance—an improvement factor between 3.5 and 4—is centered near three errors in 13-word sentences. Regardless of sentence length, no significant improvement is found after 15 errors. The worst performance reflects no improvement.

FIGURE 2. RESULTS USING THREE SIMULATED MT PACKAGES.



5. CONCLUSIONS.

Averaging ideal test sentences results in marked improvement in quality. This improvement depends on the error proportion between a group of sentences and their length. Longer sentences can generally tolerate more errors—up to approximately 15—before no further improvement occurs.

For this prototype architecture, two simplifying assumptions were made about the simulated MT package output: the sentences have similar, overlapping content, and they are all the same length. Unfortunately, actual output would be more difficult to average because natural language can correctly express equivalent ideas in numerous different ways. While the sentences might all have the same meaning, the particular words and ordering, as well as the sentence lengths, can vary greatly. Averaging sentences based solely on the words and their positions does not account for this limitation. Future work would need to introduce more flexibility into handling such problems. Performance would likely suffer, but it is expected that averaging would still show some improvement.

WYANDOT PHONOLOGY: RECOVERING THE SOUND SYSTEM OF AN EXTINCT LANGUAGE

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1. INTRODUCTION.

Wyandot is a Northern Iroquoian language, related to Seneca, Mohawk, and Cherokee, among others. While the last speakers lived in the 1960s, the last extensive fieldwork was carried out by Marius Barbeau in 1911-1912. He intended to produce the usual trio of grammar, dictionary, and texts, but did not finish. No grammar was started, and the dictionary has remained in manuscript form, probably due to the extensive misanalyses present. Only the texts were ever published, in 1960. These consisted of approximately 253 pages of legible handwritten transcription of Wyandot and typed word glosses, with an additional 51 typed pages of free translation.

To perform a phonological analysis of a language, linguists usually phonetically transcribe utterances obtained from native speakers, then search the data for contrastive and complementary distribution, seek minimal pairs, return to the speakers for verification of hypotheses, and so on. But what should be done when there are no longer any speakers of the language? How does one proceed when the linguist cannot simply go to a speaker to round out a paradigm, or run a spectrogram? This is the problem in sorting out the phonology of Wyandot.

2. WRITTEN TRANSCRIPTION.

Since there is a lack of fluent native speakers of Wyandot and, therefore, the texts cannot be retranscribed phonetically, the transcription system used by Barbeau himself will be the basis used here instead. This particular system has several drawbacks, to be discussed in the following sections. Some peculiarities can be assigned to the time period when the fieldwork was done, centering around 1912. However, this does not explain the retention of the system in publishing Barbeau (1960). Our analysis will combine standard phonological procedure, the comparative method, and historical insights. Various types of brackets will be used for different transcriptions: Barbeau's transcription will be in <angle brackets>, the phonetic realization in [square brackets], and the phonemic interpretation in /slashes/.

2.1. Barbeau's Characters. Barbeau gives a description of his transcription system in several works, with minor variations. The following description closely follows the format Barbeau used, with two exceptions. Additional notes found only in certain works are added

below the standard description, and modern IPA equivalents are placed to the side.

VOWELS

a	as in English <i>mat</i> , and in French <i>parade</i>	[æ ~ a]
	Barbeau (1915a) hedges with “a vowel closely resembling those in”	
e	as in French <i>é</i> - English <i>a</i> in <i>cave</i>	[e ~ e ^l]
ɛ	as in French <i>è</i> - English <i>e</i> in <i>pet</i>	[ɛ]
i	as in French <i>i</i> - English <i>i</i> in <i>fit</i>	[i ~ ɪ]
u	as in French <i>ou</i> , - English <i>o</i> in <i>lose</i>	[u ~ u ^w]

NASALIZED VOWELS

ɶ, ɛ̃, ɪ̃ (rare), the *a*, *e*, *i* above vowels nasalized:

ɶ	as in the French <i>marchand</i>	[ã]
ɛ̃	as in the French <i>in</i> , in <i>vin</i>	[ɛ̃]
ɶ	the open <i>o</i> nasalized as in French <i>bon</i>	[õ]

SEMIVOWELS

w	as in the English <i>wine</i>	[w]
y	as in the English <i>yes</i>	[j]

CONSONANTS

c	as in the English <i>she</i> , - French <i>chat</i>	[ʃ]
j	as in the French <i>jamaïs</i> followed by a brief <i>y</i>	[ʒ]
	Barbeau (1915a) hedges, with “closely resembling that of French <i>jamaïs</i> ” and adding that the <i>y</i> is very brief	
s	as in the English <i>sit</i>	[s]
t	as <i>t</i> with a slight aspiration	[t̚ ~ t ^h]
	Barbeau (1915a) states “approximately as in English and French”	
	Barbeau (1949) also indicates “followed by a slight aspiration”	
d	as in <i>done</i> , often preceded by a weak ⁿ	[t ~ d ~ ⁿ d]

Wyandot Phonology: Recovering the Sound System of an Extinct Language

Barbeau (1915a) hedges with “approximately”

Barbeau (1915b) refers to a “weak *n*”

k as in *key* [k ~ k^h ~ c ~ c^h]

Barbeau (1915a) states “approximately as in English”

g the sonant *g* followed by a *y*, often preceded by a weak ⁿ [g^j ~ ⁿg^j ~ ɟ ~ ɟⁿ]

Barbeau (1915a) uses *gy*, with a “preceding weak ñ”

Barbeau (1915b) also uses *gy*, described as “sonant *g* immediately followed by *y*, often with a preceding weak ⁿ, palatalized *ng* of English *sing*”

ḱ *k* followed by *y* [k^j ~ k^hj ~ c ~ c^h]

Barbeau (1915a,b) instead use *ky*

m and [m]

n as in English and French [n ~ ɲ]

ñ as the *ñ* in Spanish; the *gn* in Italian [ɲ]

r corresponding to the English *r* [ɹ]

Barbeau (1915b) hedges with “roughly”

ɾ rare; deep palatal, tending to disappear ?

Barbeau (1915a, b) lack this character

h aspiration always followed by a vowel [h]

Superscript letters indicate very brief, and sometimes unvoiced consonants and vowels, as in

ⁿdătrá^askwĩjù^ʰⁿdɪ̃

DIACRITICAL MARKS

ʰ glottal stop or catch as in ⁿgá^ʰwic [ʔ]

˘ breathing after a vowel and before a consonant as in a˘cɛ˘k [h]

ˈ over a vowel shows the main stress or accent in a word; it usually corresponds to a rising pitch of the voice

Barbeau (1915b) refers to this as “high pitch”

ˌ minor or weaker accent

Barbeau (1915a) uses “secondary”

- a raised period after a vowel indicates that it is long, as in $\text{ĩy}^{\circ}\text{t}^{\circ}\text{e}^{\circ}$
- ˘ over a vowel makes it brief: $\text{t}^{\circ}\text{e}^{\circ}\text{h}^{\circ}\text{a}^{\circ}\text{t}$

Unmarked vowels are of medium length. Two consecutive brief vowels may be combined into one main accent: $\text{ũr}^{\circ}\text{e}^{\circ}\text{h}^{\circ}\text{a}^{\circ}$ (N.B. the two ˘ are linked above by $\sim$, with a ˘ above the link [CK])

Difficulties in ascertaining the phonetic nature of the Barbeau characters are readily apparent. For instance, vowels are often described as like equivalents in French and English, even though French and English vowels are often quite different phonetically. The offglides on English $[\text{e}^{\text{h}} \text{u}^{\text{w}}]$ do not appear on French $[\text{e} \text{u}]$, for one example. Also, although English-speaking beginning students of French may often perceive French $[\text{a}]$ as $[\text{æ}]$, the two are distinct. Although the nasal vowels are described as nasalized versions of the oral vowels, the descriptions of oral $\langle \text{a} \rangle$ as $[\text{æ} \sim \text{a}]$ do not correspond to the nasal $\langle \text{a} \rangle$ as $[\tilde{\text{a}}]$.

At times Barbeau describes $\langle \text{t} \text{ k} \rangle$ as “unaspirated surds”, while at other times $\langle \text{t} \rangle$ is contradictorily described as having a “slight aspiration”. Further, $\langle \text{t} \text{ n} \rangle$ described as similar to English and French obscures the actual place of articulation, which in English is alveolar but in French is dental. The “deep palatal” $\langle \text{ɾ} \rangle$ is not given an IPA equivalent, since it is not clear what Barbeau meant by the description.

Additional characters not mentioned but used include:

- $^{\text{m}}$ in Barbeau (1915a), appearing in the environment $\text{u}^{\text{m}}\text{w}$
- $^{\text{n}}$ in Barbeau (1915a), appearing in the environment $\text{ĩ}^{\text{n}}\text{gy}$
- $^{\text{ŋ}}$ in Barbeau (1915b), appearing in the environment $\text{ĩ}^{\text{ŋ}}\text{gy}$
- ç rarely used, in Barbeau (1960)
- $^{\text{w}}$ rarely used, in Barbeau (1960)
- $^{\text{y}}$ rarely used, in Barbeau (1960)

The characters $\langle ^{\text{m}} \text{ } ^{\text{n}} \text{ } ^{\text{ŋ}} \rangle$ are apparently $\langle ^{\text{n}} \rangle$ assimilating to a labial or velar environment. On the analogy $\text{k}:\text{k}::\text{c}:\text{ç}$, $\langle \text{ç} \rangle$ is presumably equivalent to $\langle \text{cy} \rangle$. The “superior letters” refer primarily to $\langle ^{\text{n}} \rangle$, which usually appears before d or after a nasal vowel, as well as to echoed vowels after glottal stops, i.e., $\text{ĩ} \text{ e } \text{ẽ} \text{ ɛ } \text{ɛ̃} \text{ a } \text{ã} \text{ o } \text{õ} \text{ u}$.

Some of the variations can be explained according to Barbeau's personal notes, *inter alia*, as character alternations based on typesetting demands. For typesetters unable to create $\langle \text{k} \rangle$, Barbeau would use $\langle \text{ky} \rangle$, for those unable to print $\langle \text{é} \rangle$ he would use $\langle \text{e}' \rangle$, and so on.

3. OVER-DIFFERENTIATION.

In addition to the previously described peculiarities, the Barbeau system has several other drawbacks. One of these is over-differentiation. For instance, the same word may appear with several different length, stress, and nasalization patterns that do not contrast, as in (1):¹

- (1)
- a. ǎh̃h̃ǎq̣ *he said* TN:02:068:03
 - b. ah̃h̃ah̃q̣ *he said* TN:12:113:18
 - c. ah̃h̃h̃ǎq̣ *he said* TN:01:062:20; TN:02:063:36
 - d. ǎh̃h̃ah̃q̣ *he said* TN:12:114:15
 - e. ǎh̃h̃h̃ǎq̣ *he said* TN:01:060:09
 - f. ah̃h̃h̃ah̃q̣ *he said* TN:12:114:58
 - g. $\text{ah̃h̃}^n\text{h̃ǎq̣}$ *he said* TN:01:060:03

Each of these examples shows a different pattern, as demonstrated in (2):

- (2)
- a. $\begin{matrix} \sim & \sim & / \\ & \backslash & / \\ & \sim & \end{matrix}$
 - b. $\begin{matrix} & \backslash & / \\ & \sim & \end{matrix}$
 - c. $\begin{matrix} & \backslash & / \\ & \sim & \end{matrix}$
 - d. $\begin{matrix} \sim & \backslash & / \\ \sim & \backslash & / \\ \sim & \backslash & / \end{matrix}$
 - e. $\begin{matrix} & \backslash & / \\ & \sim & \end{matrix}$
 - f. $\begin{matrix} \sim & \backslash & / \\ \sim & \backslash & / \\ \sim & \backslash & / \end{matrix}$
 - g. $\begin{matrix} & \backslash & / \\ & \sim & \end{matrix}$

Note that the only vowel that is consistent is the last. All of the others vary in length, stress, and nasalization.

Vowels may also appear with different qualities, which do not contrast, as can be seen in (3):

- (3)
- a. $\text{a}^{\text{a}}\text{wé}^{\text{h}}\text{q̣}$ *she thought* TN:04:082:24
 - b. $\text{a}^{\text{a}}\text{wé}^{\text{h}}\text{q̣}$ *she thought* TN:01:059:14; TN:04:079:41

However, these two sounds cannot simply be considered allophones, since <ε> enters into alternations that <e> does not. <ε> alternates with <ξ> as in (4); however, <e> and <ξ> not alternate:

- (4)
- a. $\text{yǎñé}^{\text{n}}\text{q̣}$ *dog* TN:16:127:62
 - b. $\text{yǎñξ}^{\text{n}}\text{q̣}$ *dog* TN:16:126:36

This time there is a noncontrastive difference between <ɛ> and <ɛ̃>.

Any phonemicization of Barbeau's materials must take into consideration that not all of his characters represent distinct phonemes, but rather may indicate minor variations in sound.

3.1. Under-differentiation. Another problem with the Barbeau orthography is that it is under-differentiated. That is, not all distinctive contrasts are marked.

All other Iroquoian languages contrast the simple phonemes /t/ and /k/ with the clusters /th/ and /kh/. Example (5) shows two forms based on the Cayuga root *-ehst-* *use* (Mithun and Henry 1982). (Underlining signifies devoicing in Cayuga orthography.)

- (5) a. tesɛnʔqá-ehst-qh *you are using your head*
 b. s-ehst-haʔ *you use it*

The first form adjoins t and q across a morpheme boundary, while the second has t and h. These are usually phonetically realized as [t k] and [tʰ kʰ]. They are frequently represented as <d g> and <t k> in orthographies developed by English speakers or intended for English-speaking learners.

Neither stop-h cluster appears in the Barbeau texts. Either the transcription system fails to represent the distinction between /t k/ and /th kh/, or Wyandot is the only Iroquoian language to have lost the distinction. Knowing that <d g> both appear in the texts, it may be hypothesized that the Barbeau orthography is English-based. However, <d> is cognate with /n/, not /t/, in the other languages. Compare in (6) forms based on the root for *drum/barrel* in Wyandot (6a) and Seneca (6b) (Chafe 1967):

- (6) a. hʉⁿ-^ʔnda'kw-ǎ'éhǎkeʔ *he the drum goes on beating* TN:03:076:09
 b. ye-ʔnqhkwa-aʔésthaʔ *drumstick*

Here the initial ʔ of the morpheme as well as the final hkw, are the same in both languages. The q in Seneca comes from the rule in (7):

- (7) a → q / n _

This leaves Seneca /n/ cognate with Wyandot <ⁿd>. Therefore, Barbeau did not interpret unaspirated [t] as an English speaker would, as /d/, and thus did not similarly misinterpret an aspirated [th] cluster in English terms, as /t/.

Simple English-biased misrepresentations of [kh] and [k] in a parallel manner as [k] and [g] are not the proper explanation either, as [g] is an allophone of /d/ (see 4.1).

Furthermore, sometimes Wyandot <t> corresponds to /t/ in cognate languages, and sometimes to /th/. In (8) compare the root <hati> *they* in Wyandot (8a) and **-hati-** *they* in Cayuga (8b):

- (8) a. **ă-hàti-** 'cróⁿga' *they make* TN:07:100:40
 b. **hati-**?trehtó:nihs *they make cars*

Here the Wyandot stop is equivalent to a Cayuga simple stop. But sometimes Wyandot <t> has /th/ correspondences. Compare <te^ht> *pound* in Wyandot (9a) with **-the^ht-** *grind; pound* in Cayuga (9b):

- (9) a. **săí-**te^ht *again they (2) pound corn* TN:04:083:09
 b. **akate-**the^ht-hró?kt?aqh *I am out of flour*

or Wyandot <ratε> (10a) and Cayuga **-rathε-** (10b) *climb*:

- (10) a. **hă-rátε-** he *climbs* TN:13:118:20
 b. **ha^ht-rát:thε-h** *climb up there!*

In each of these cases Wyandot <t> corresponds to Cayuga /th/. The same can be shown for Wyandot <k> and /k kh/ cognates in Mohawk. Example (11) compares Wyandot <gak> *marry* (11a) and Mohawk **-nyak-** *get married* (11b) (Mohawk from Michelson 1973).

- (11) a. **ahăhu-gák-**a^h *he got married* TN:18:134:29
 b. **waké-nyak-s** *I get married*

Note that here Wyandot <k> is equivalent to Mohawk /k/. In the following example, Wyandot <tsi^hke^ht> (12a) is contrasted with Mohawk **-tsikhe^ht-** (12b) *sugar*.

- (12) a. **du-tsí^hke^ht-**a^h *in a sugar lump* TN:14:123:20
 b. **o-tsikhè:t-a** *sugar*

As can be seen, this equivalency is between <k> and /kh/. Thus, simple stops in Barbeau's orthography sometimes represent simple stops, and sometimes represent aspirated clusters, based on comparison to related languages.

Thus any phonemicization of Barbeau's materials must take into consideration that not all of his characters represent individual phonemes, but rather may indicate clusters of distinct phonemes.

4. CONSONANTS.

The various consonants Barbeau uses are: <t k ʔ ⁿt ⁿk ⁿd g ⁿd ⁿg s c ɕ h ʕ j m ^mn ⁿ ^ɲ ñ ^ɲr ɾ w ^w y ^y >.

Since <k ɕ ñ> represent clusters, they can be eliminated in favor of those corresponding clusters: <ky cy ny>. The description of <g> always refers to an offglide, so this is better represented as <gy>. This reduces the inventory to <t k ʔ ⁿt ⁿk d g ⁿd ⁿg s c h ʕ j m ^mn ⁿ ^ɲ r ɾ w ^w y ^y >.

4.1. Allophones. Further reduction can be obtained by eliminating allophones. The voiced stops are in free variation with prenasalized voiced stops. Thus, <d> and <ⁿd> are in free variation, as in (13), and <g> and <ⁿg> are, as in (14).

- (13) a. tayēmḡ·-d-úrḡhəʔ *thou my desire findest out* TN:04:088:18²
 b. ähämḡ·-ⁿd-úrḡhəʔ *he her desire finds out* TN:04:088:47

- (14) a. ähì·-g-áhə *they (2) eat* TN:04:081:34
 b. ähäti-ⁿg-áhəʔ *(for) them to eat* TN:03:076:28

Since <dy> does not occur, and <g> is described as followed by y, we can eliminate <g> through the rule <d> → <g> / _ [y]. Furthermore, as can be seen in (15-16), <g> is cognate with /ni ~ ny/ elsewhere in Iroquoian.

- (15) a. ähá-ʕcr-ḡⁿg-aʔ *he makes* TN:01:062:06
 b. ase-hsr-ḡⁿi-ʔ *you made it*
 c. k-ḡⁿi *I make*
- (16) a. äñḡ·-gák-a *I you married* TN:02:068:14
 b. a:yqkhní:-nya:k *we two would marry*
 c. waké-nyak-s *I get married*

The root for *make* ends in -ni- in Cayuga (15b) and Mohawk (15c), but <g> in Wyandot (15a). *Marry* begins with -ny- in Cayuga (16b) and Mohawk (16c) but <g> in Wyandot (16a).

Most importantly, roots can be found within Wyandot which show <d ~ g> alternation:

- (17) a. yà·-ré·ⁿd-a' *trap* TN:05:091:29
 b. hă·-rĕⁿd-ăř·ñq's *he his traps sets several* TN:16:126:23
 c. ahăyă·-rĕⁿg-a' *he trap takes away* TN:05:091:24

In (17) the root for *trap*, <rĕⁿd>, varies between final <d> in (17a-b) and final <g> in (17c).

The prenasalized voiceless stops are in variation with the simple versions, appearing after a previous nasal segment.

- (18) a. ăhu·nq-ⁿt *she (to) him gave* TN:15:125:58
 b. ăyq·nq-t *will I (to) you give* TN:15:125:43

Here two forms of *give* show free variation of <ⁿt> and <t> after a nasal <q>. In (19) <ⁿk> varies with <k> after a nasal.

- (19) a. da'yăĕ-ⁿk-wa' *that she planted (the seeds)* TN:01:061:11
 b. hĩñĕ-k-wa' *they two are to plant* TN:04:090:30

A rule of the form <r> → <ř> / _ <h> eliminates the “deep palatal”:

- (20) a. dă'ú·-řh-ĕhă' *the next day* TN:04:088:05
 b. dĭyá·-řh-ĭ' *that tree* TN:03:074:28

Almost all instances of <m> can be derived from a <w> in the environment of a nasal segment. <w> does not appear adjacent to a nasal, and <m> does not appear without another nasal. This can be covered by the rule <w> → <m> / () % Ỳ. (Where % indicates either a preceding or following environment.)

- (21) a. ăhă-mĕ-ⁿdúrĕhă' *he her desire finds out* TN:04:088:47
 b. ăhătikaskà^ah-ŏm-a' *they a division made* TN:07:098:48
 c. ahăyqñ-òⁿm-a'cĕⁿdré· *he their bills tied up* TN:11:110:16

However, a few unexplained instances have <m> adjacent to <ă>, which itself is not phonemic (see 5.1).

- (22) mă-hă'cănq· *most so* TN:02:064:05

These examples are almost all particles, however, and it is known that particles can have phonological peculiarities (Mithun 1982).

Since <ʰ> is “breathing after a vowel and before a consonant” and <h> is “aspiration always followed by a vowel”, it is clear that they are allographs, with <h> used where English has /h/, and <ʰ> used where English does not allow /h/.

As mentioned previously, <^{m̥n̥}> are assimilations of <^{n̥}> to labial and velar environments. <^{n̥}> itself almost only occurs in the prenasalized stops, which have already been dealt with. The exceptions occur after nasal vowels, as in (23):

- (23) a. ah-^ñ-hăqʰ *he said* TN:01:060:03
 b. ku^ug-^ñ-seʰ *snake* TN:08:102:18

These allophones, allographs, and other variations reduce the set of consonants to the following: <t k ʰ d s c h j n r w y>.

5. VOWELS.

Barbeau uses 62 vowel characters, plus an additional 10 raised characters: <ⁱ ^e ^ɛ ^ɛ ^ɛ ^a ^ə ^o ^ɔ ^u>. The 62 are shown below:

i	í	ì	ĩ	í	ì		í	ì			
e	é	è	ě	é	è	ɛ	é	è	ě	é	è
ɛ	é	è	ě	é	è	ɛ	é	è	ě	é	
a	á	à	ă	á	à	ə	á	à	ă	á	
o	ó	ò	õ	ó	ò	ɔ	ó	ò	õ	ó	ò
u	ú	ù	ũ	ú		u	ú	ù			

The diacritic patterns can be shown in matrix format for clarity.

	plain	ˈ	ˊ	ˋ	ˌ	ˍ	ˎ	ˏ	ː	ˑ	˒
i	+	+	+	+	+	+	-	+	+	-	-
e	+	+	+	+	+	+	+	+	+	+	+
ɛ	+	+	+	+	+	+	+	+	+	+	-
a	+	+	+	+	+	+	+	+	+	+	-
o	+	+	+	+	+	+	+	+	+	+	+
u	+	+	+	+	+	-	+	+	+	-	-

5.1. Allophones. These 72 characters can be reduced by eliminating allophones. <e> and <ɛ> alternate optionally.

- (24) a. ǎhàti-jé·rat *they used* TN:07:099:05
b. àhǎ-éraʔt *he used* TN:13:119:31

In (24) the root <(y)eraʔt> *use* alternates between <e> and <ɛ>.

In a parallel fashion, the nasal equivalents <ɐ> and <ɛ̃> also optionally alternate.

- (25) a. ǎhatihàʔc-ǎ-taʔ *they a council held* TN:07:098:42
b. hǎtēháʔc-ɛ-ʔt *they a council hold* TN:09:104:32

This shows a vowel quality alternation in the root <haʔcɐt> *hold council*.

Unfortunately, there is also a nasal-oral alternation between <ɛ̃> and <ɛ>, though this is in the vicinity of another nasal segment.

- (26) a. aʔtēm-ǎ-tayeʔ *every day* TN:01:059:03
b. aʔtēm-ɛ-ʔtayeʔ *every day* TN:12:112:23

In (26) <ɛ̃> and <ɛ> are in free variation after nasal <m>. Some or all of these may be due to errors in transcription. That is, the nasal nature of a nearby nasal segment might have masked nasalization on the vowel itself. These variations may thus actually be English-biased perception mistakes, which would also account for the several other nasal-oral vowel alternations <i ~ ɨ>, <a ~ ǎ>, <u ~ ʊ>, and <o ~ ɔ>.

The character <ɨ> only appears next to a nasal, and <i> does not appear next to nasals. Thus, the alternation of <i> and <ɨ> is the result of nasalization from another nasal segment.

- (27) a. d-ǝ̀-ⁿdǎwé^{et} *we two are brother and sister in law* TN:04:084:14
 b. n-ǝ̀i-ⁿdǎwé^{et} *we two are brothers and sister in law* TN:04:084:23

Here the alternation of <j> with <i> can be explained through the presence of the preceding nasal <ǝ>.

The same holds for <ǎ>, except that a laryngeal <h^ʔ> can intervene.

- (28) a. ǎh-ǎ-ⁿg-ǎ^ʔ *he her chases (after)* TN:02:069:26
 b. ǎhǎtikaskà^ʔh-ǝ̀mǎ-^ʔ *they a division made* TN:07:098:48
 c. ahǎjá^ʔtür-ǝ̀hǎ-^ʔ *he me finds* TN:01:061:27

Example (28a) shows <ǎ> before a nasal <ⁿg> while (28b) shows <ǎ> after nasals <ǝm>. (28c) shows nasal assimilation across an intervening laryngeal <h>.

As mentioned previously, sometimes <ǎ> appears without a nasal environment, especially in particles.

- (29) hǎ^ʔra^ʔ *only* TN:04:081:37; TN:04:087:48; TN:04:088:14

<u> can appear as nasalized <ǘ> in the vicinity of a nasal, in that oral <u> does not occur next to a nasal, and <ǘ> only appears adjacent to a nasal. Example (30) shows the occurrence of <ǘ> adjacent to a nasal.

- (30) t-ǘ^ʔd-i *also* TN:02:065:27

The opposite occurs with <ǝ> and <o>, where <o> only appears adjacent to a nasal. That is, where <ǎ ǘ> are nasal allophones of <a u>, <o> is an allophone of <ǝ>.

- (31) a. kasǎká^ʔke-nǝ *here thou liest down* TN:04:086:09-10
 b. ǎhǎká^ʔkě-no *he lies down* TN:12:112:42

In (31) we see an alternation of <ǝ> and <o> following <n>. As with <ε> as allophone of <ǝ>, this may be a transcription error, the nasality of a nearby nasal segment hiding the nasality of the vowel <ǝ>.

Finally, vowels may be echoed optionally after a glottal stop.

- (32) a. yăhà·w – Ǿⁱ *she carries* TN:04:090:26
b. yǾⁿtẵr – é^e *lake* TN:08:103:50
c. ăhăt – Ǿ[·] – wa[·] *he runs off* TN:11:109:13
d. y – Ǿ[·] – ric *lion* TN:09:105:33
e. ʔe[·]skēmǿⁿdẵr – á^a – cǾñǾ[·] *will thou (with) me converse* TN:04:079:10
f. ŭwa[·]h – Ǿ[·] – rǾ[·] *another one* TN:03:075:08
g. c – Ǿ[·] – mǾ[·] *by far* TN:04:080:49
h. yǿ[·]tij – ú^u – rŭn – Ǿ[·] *prairie turtle tribe* TN:09:105:13

This example shows each of the echoed vowels appearing after a glottal stop. The nasal-oral alternations discussed above also occur amongst the echoed vowels, as can be seen in (32g), where oral <Ǿ> appears with nasal <Ǿ̃>.

6. PHONEMES.

With the various reductions and alternations in sections 4 and 5, and modernizing the symbols to a traditional Americanist set, we can obtain the phonemic inventory of Wyandot. The consonant phonemes are listed below:

(33) Consonant Phonemes

	labial	alveolar	palatal	velar	glottal
voiceless stop		t		k	ʔ
voiced stop		d			
voiceless fricative		s	ʃ		h
voiced fricative			ʒ		
nasal		n			
resonant		r			
glide	w		y		

The term “alveolar” is used here with the caveat that Barbeau was not explicit in his description of the place of articulation, as he described it as similar to both English (and thus alveolar), and to French (and thus dental).

There is a single voiced stop, /d/, contrasting with a voiceless stop /t/. This is interesting in that no other Iroquoian language has a voicing distinction. Furthermore, another unusual feature is the presence of a voiced fricative /ʒ/. This is again counter to Iroquoian languages’

lack of a phonemic voicing distinction. It is also unusual cross-linguistically, with a /š ž/ contrast without a corresponding /s *z/ contrast.

Phonotactics are more problematic, in that the aspirated clusters /th kh/ are unmarked in the Barbeau orthography. However, we can assess consonant clusters aside from these. The initial consonant clusters (#CC) include: ts-, tr-, kw-, ky-, dr-, dy-, st-, sk-, šr-, šy- and ny-, as shown in the following table:

(34) #CC Clusters

	t	k	s	r	w	y
t	-	-	+	+	-	-
k	-	-	-	-	+	+
d	-	-	-	+	-	+
s	+	+	-	-	-	-
š	-	-	-	+	-	+
n	-	-	-	-	-	+

There is a single #CCC cluster, skw-.

Final clusters (CC#) can consist of only two members: -ʔt, -ʔk, -ʔs, -st, -ht, -hk, -hs, -hš, -yh, and -rh, as shown:

(35) CC# Clusters

	t	k	s	š	h
ʔ	+	+	+	-	-
s	+	-	-	-	-
h	+	+	+	+	-
y	-	-	-	-	+
r	-	-	-	-	+

Medial clusters (-CC-) can of course be longer. The medial CC clusters include: -tr-, -kw-, -ky-, -ʔt-, -ʔk-, -ʔd-, -ʔs-, -ʔh-, -ʔž-, -ʔr-, -ʔw-, -ʔy-, -dr-, -dy-, -st-, -sk-, -sh-, -šk-, -šr-, -ht-, -hk-, -hs-, -hš-, -hr-, -ny-, -rh-, -yʔ-, and -yr-

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(36) -CC- Clusters

	t	k	ʔ	d	s	š	h	ž	r	w	y
t	-	-	-	-	-	-	-	-	+	-	-
k	-	-	-	-	-	-	-	-	-	+	+
ʔ	+	+	-	+	+	-	+	+	+	+	+
d	-	-	-	-	-	-	-	-	+	-	+
s	+	+	-	-	-	-	+	-	-	-	-
š	-	+	-	-	-	-	-	-	+	-	-
h	+	+	-	-	+	+	-	-	+	-	-
n	-	-	-	-	-	-	-	-	-	-	+
r	-	-	-	-	-	-	+	-	-	-	-
y	-	-	+	-	-	-	-	-	+	-	-

Medial CCC clusters are: -ʔts-, -ʔtr-, -ʔkw-, -ʔky-, -ʔdr-, -ʔdy-, -ʔsk-, -ʔny-, -skw-, -sky-, -hts-, -hkw-, -hky-, -hst-, -hsk-, -hšt-, -hšr-, -hšy-, and -hny-. Medial CCCC clusters are: -ʔskw-, -hstr-, -hskw-, and -hsky-.

The Wyandot vowel phoneme inventory can likewise be listed.

(37) Vowel Phonemes

i u
e
ɛ ɔ
a

This can be further rearranged if desired into a symmetrical shape:

(38) Vowel Phonemes

i u
ɛ ɔ
e a

Although Barbeau described the nasal phonemes as /ɛ ɔ/, for simplicity they can be written /ɛ ɔ/. Few vowel clusters can be found. These are: ae-, ai-, aɔ-, eu- initially; -VeV- medially; -ae, -aɔ, and -eu finally.

On a final note, there are various rules whose synchronic or diachronic nature cannot be determined. Some dialects of various Iroquoian languages have a rule $t \rightarrow k / _ y$. It appears this rule also is at work in Wyandot, since there are occurrences of *ky* in Wyandot but no **ty*. However, most examples are not adequate to decide whether the rule is synchronic or diachronic. Fortunately, there are a small number of examples of alternations between *t* and *k* before *y*.

- (39) a. $y\ddot{a}r\acute{o}\cdot ta^{\flat}$ *log* TN:11:109:36
 b. $ah\ddot{a}r\acute{o}^{\flat}ka^{\flat a}$ *he log down* TN:19:143:47

This example shows an alternation in the root **-rqt-** *log* between final /k/ before /y/ (shown by the <ḱ>), and final /t/ elsewhere.

- (40) a. $h\ddot{u}n\acute{x}r\acute{o}\cdot ti^{\flat}$ *he goes out hunting* TN:04:087:22
 b. $ah\ddot{u}n\acute{x}r\acute{o}ke^{\flat}s$ *he goes out hunting* TN:12:112:19

The root for *hunt*, **-nḗrqt-**, appears with final /t/ except in (40b), where it appears as /k/ before the following /y/.

Thus, there is a phonemic rule changing a voiceless stop from alveolar to velar before *y*, and a phonetic rule changing a voiced stop from alveolar to velar before *y*. The phonemic rule appears to have been extended in Wyandot to cover /w/ in addition to /y/, that is, to both glides. This can be seen comparing **-kahkw-** *see* in Wyandot (41a) with **-kaht-hw-** *look at, see* in Seneca (41b):

- (41) a. $\acute{a}h\acute{a}-ka^{\flat}kw-a^{\flat}$ *he saw* TN:12:113:01
 b. $ak\acute{a}t-kathw-\phi h$ *I've looked at it*³

The Seneca form shows /thw/, while the Wyandot has /kw/. The cluster /tw/ does not occur in Wyandot. And the only example of <gw> is cognate with /kw/ in the other languages. There is not enough evidence to declare the rule synchronic. Thus, we have three very similar rules:

- (42) a. $t \rightarrow k / _ y$
 b. $/d/ \rightarrow [g] / _ y$
 c. $(*)t > k / _ w$

These could be written as one if we mixed phonetic and phonemic notation, and diachronic and synchronic changes.